Strong Nullstellensatz via Rabinowitsch's Trick

An End-to-End Meta-Math Walkthrough

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Abstract

Turn the assertion that a polynomial never becomes nonzero on the variety into an inconsistent system by adjoining an inverse variable. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

AG-03 / Algebraic Geometry / Report v1

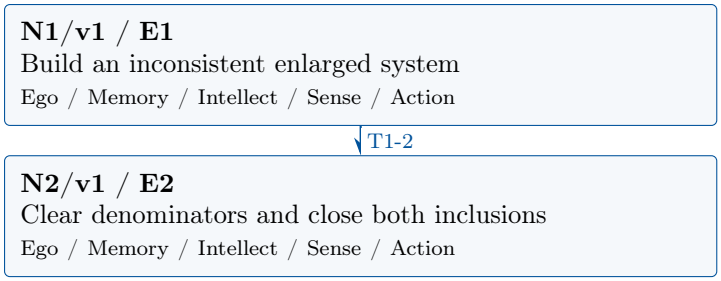
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For an algebraically closed field k, finite n, and any ideal $I \subset k[x_1, \ldots, x_n]$, prove $I(V(I)) = \sqrt{I}$. Here $I(S)$ is the ideal of polynomials vanishing on S, and $\sqrt{I} = \{f : f^m \in I \text{ for some integer } m \geq 1\}$.*

A. Foundations and permitted imports

Import AG-02 in the following exact form: a proper ideal in a finite-variable polynomial ring over algebraically closed k has a common zero. Use localization at the powers of a nonzero polynomial and finite ideal-combination identities. No effective bound on the exponent is assumed or concluded.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $1 \in J$, so $1 = \sum_{j=1}^{s} A_j(x,y)g_j(x) + B(x,y)(1 - yf(x))$ for finitely many $g_j \in I$.

Receiving activity. N2.1

Compatibility. E1/v1; $f \neq 0$ for the localization step, with $f = 0$ treated separately. Check: Zero cannot be inverted, and the exponent clearing all denominators must be chosen for the finite entire sum.

III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

A. N1/v1: Build an inconsistent enlarged system

Object and role. Combine vanishing of f on the original zeros with the equation that would make it invertible. **Local obligation:** $1 \in J$, so $1 = \sum_{j=1}^{s} A_j(x,y)g_j(x) + B(x,y)(1 - yf(x))$ for finitely many $g_j \in I$.

Inputs. $f \in I(V(I))$ and the ideal $J = (I, 1 - yf)$ in $k[x_1, \ldots, x_n, y]$.

E — Ego

Publish E1/v1 only after establishing this obligation: $1 \in J$, so $1 = \sum_{j=1}^{s} A_j(x,y)g_j(x) + B(x,y)(1 - yf(x))$ for finitely many $g_j \in I$.

M — Memory

$f \in I(V(I))$ and the ideal $J = (I, 1 - yf)$ in $k[x_1, \ldots, x_n, y]$. Retain the stated assumptions and import versions.

I — Intellect

Combine vanishing of f on the original zeros with the equation that would make it invertible.

S — Sense

The new variable y is independent; the weak Nullstellensatz applies in $n + 1$ variables over the same algebraically closed field.

A — Action

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 A common zero (a, b) of J would satisfy $a \in V(I)$, hence $f(a) = 0$, while also $1 - bf(a) = 0$. This would give $1 = 0$, so J has no common zero.

N1.2 By the contrapositive of AG-02, J is the whole ring. Membership of 1 in the generated ideal means exactly a finite polynomial-combination identity of the displayed form; finite generation of all of I is not required for this individual identity.

Export E1/v1. $1 \in J$, so $1 = \sum_{j=1}^{s} A_j(x,y)g_j(x) + B(x,y)(1 - yf(x))$ for finitely many $g_j \in I$.

Acceptance gate. The new variable y is independent; the weak Nullstellensatz applies in $n + 1$ variables over the same algebraically closed field.

Targeted repair. If inconsistency is equated with the unit ideal over an arbitrary field, restore the algebraic-closure hypothesis before invoking AG-02. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Clear denominators and close both inclusions

Object and role. Substitute $y = f^{-1}$, multiply by a sufficiently large power of f, and prove the converse directly. **Local obligation:** $I(V(I)) = \sqrt{I}$.

Inputs. E1/v1; $f \neq 0$ for the localization step, with $f = 0$ treated separately. Incoming proof exports: E1/v1 at N2.1.

E — Ego

Publish E2/v1 only after establishing this obligation: $I(V(I)) = \sqrt{I}$.

M — Memory

E1/v1; $f \neq 0$ for the localization step, with $f = 0$ treated separately. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

I — Intellect

Substitute $y = f^{-1}$, multiply by a sufficiently large power of f, and prove the converse directly.

S — Sense

Zero cannot be inverted, and the exponent clearing all denominators must be chosen for the finite entire sum.

A — Action

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 If $f = 0$, it is already in $\sqrt{I}$. Otherwise evaluate the E1 identity in $k[x_1, \ldots, x_n, 1/f]$ at $y = 1/f$. The last term vanishes, leaving $1 = \sum_j A_j(x, 1/f) g_j(x)$.

N2.2 Choose $m \geq 1$ at least the largest y-degree of the finitely many A_j. Multiplication by f^m yields $f^m = \sum_j h_j(x) g_j(x)$ with polynomial h_j, so $f \in \sqrt{I}$. Conversely, if $f^m \in I$ and $a \in V(I)$, then $f(a)^m = 0$ in the field k, hence $f(a) = 0$. This proves both inclusions, including I equal to the unit ideal.

Export E2/v1. $I(V(I)) = \sqrt{I}$.

Acceptance gate. Zero cannot be inverted, and the exponent clearing all denominators must be chosen for the finite entire sum.

Targeted repair. If the conclusion is $f \in I$ rather than $f^m \in I$, retain the denominator-clearing exponent; discarding it loses multiplicity information. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Build an inconsistent enlarged system

1. **N1.1.** A common zero (a, b) of J would satisfy $a \in V(I)$, hence $f(a) = 0$, while also $1 - bf(a) = 0$. This would give $1 = 0$, so J has no common zero.

2. **N1.2.** By the contrapositive of AG-02, J is the whole ring. Membership of 1 in the generated ideal means exactly a finite polynomial-combination identity of the displayed form; finite generation of all of I is not required for this individual identity.

B. N2 — Clear denominators and close both inclusions

1. **N2.1.** If $f = 0$, it is already in $\sqrt{I}$. Otherwise evaluate the E1 identity in $k[x_1, \ldots, x_n, 1/f]$ at $y = 1/f$. The last term vanishes, leaving $1 = \sum_j A_j(x, 1/f) g_j(x)$.

2. **N2.2.** Choose $m \geq 1$ at least the largest y-degree of the finitely many A_j. Multiplication by f^m yields $f^m = \sum_j h_j(x) g_j(x)$ with polynomial h_j, so $f \in \sqrt{I}$. Conversely, if $f^m \in I$ and $a \in V(I)$, then $f(a)^m = 0$ in the field k, hence $f(a) = 0$. This proves both inclusions, including I equal to the unit ideal.

□

V. WORKED ORCHESTRATION

For $I = (x^2, y)$ in $\mathbb{C}[x, y]$, $V(I) = \{(0, 0)\}$ and $I(V(I)) = (x, y)$. The polynomial x is not in I, but $x^2 \in I$. The enlarged identity for $f = x$ is $1 = (1 + tx)(1 - tx) + t^2x^2$, illustrating exactly how substitution $t = 1/x$ recovers the square.

A. Boundary and hypothesis audit

Over $\mathbb{R}$, $I = (x^2 + 1)$ has empty real zero set, so $I(V(I)) = \mathbb{R}[x]$ while $\sqrt{I} = I$. Over an algebraically closed field the theorem concerns radicals, not equality with arbitrary nonradical ideals.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N2 export: every polynomial vanishing on $V(I)$ belongs to I. The worked example $x \notin (x^2, y)$ refutes it. **Diagnosis:** Action discarded the power introduced when clearing denominators. **Repair:** export $f^m \in I$ and hence $f \in \sqrt{I}$. This narrows the rejected overclaim to the theorem actually stated in §1; the proof there already includes the correct exponent.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Deduce the inclusion-reversing correspondence between affine algebraic sets and radical ideals; then compare reduced coordinate rings with nonreduced schemes.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. J. S. Milne, Algebraic Geometry, chapters on commutative algebra, algebraic sets, and affine varieties; author's mathematics site.